1. Rapid Software Development

- **Goal**: Prioritize rapid development and delivery to respond quickly to changes in business requirements.
- **Importance**: Traditional plan-driven methods often fail to adapt to fast-changing business environments. Agile methodologies aim to address this by focusing on evolving needs rather than a fixed plan.
- **Approach**: Agile splits the development process into small, manageable increments, each developed quickly and evaluated by stakeholders.

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2. Agile Development Approach

- **Iterative Process**: Unlike linear development models, agile interleaves specification, design, and implementation. This allows for changes in requirements at any stage of development.
- **Incremental Delivery**: Software is delivered in iterations or "versions," with stakeholder involvement in defining and evaluating each version.
- **Minimal Documentation**: The focus is on delivering functional software rather than extensive documentation. Agile uses tools to automate processes, like testing, to save time.

Key Differences:

- **Plan-Driven vs. Agile**: In plan-driven approaches, stages (e.g., requirements, design, testing) are separated. In Agile, these stages are combined, enabling flexibility.

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3. Agile Manifesto and Principles

- **Core Values**:
    1. **Individuals and Interactions** over processes and tools.
    2. **Working Software** over comprehensive documentation.

3. **Customer Collaboration** over contract negotiation.
4. **Responding to Change** over following a fixed plan.

- **12 Principles** of Agile include:
  - Customer satisfaction through early and continuous delivery.
  - Welcoming changing requirements, even late in the project.
  - Delivering working software frequently (weeks rather than months).
  - Daily collaboration between business and technical teams.
  - Sustainable development pace for long-term effectiveness.

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4. Extreme Programming (XP)

- **Overview**: XP is an influential agile methodology focused on improving software quality and responsiveness. It includes practices designed to keep code simple and adaptive.
- **Practices in XP**:
  - **User Stories**: Users create short narratives to define software requirements. The team then breaks these into tasks and prioritizes them for each release.
  - **Refactoring**: Continuously improving code structure without altering functionality, which simplifies future modifications.
  - **Test-First Development**: Writing tests before the code, ensuring each new functionality is tested immediately.
  - **Pair Programming**: Two programmers work together on the same code, promoting knowledge sharing and reducing errors.

Diagram:

- **XP Release Cycle**: Diagram typically includes a loop showing continuous testing, coding, and refactoring phases for iterative improvements.

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5. Scrum Framework

- **Phases**:

    - **Planning Phase**: Outline project objectives and define high-level requirements.
    - **Sprint Cycles**: Each sprint (typically 2-4 weeks) is a focused development cycle producing a working product increment.
    - **Closure Phase**: Finalize documentation and conduct a project review.
- **Roles**:
    - **Product Owner**: Manages the product backlog, prioritizes tasks, and ensures requirements align with customer needs.
    - **Scrum Master**: Facilitates Scrum meetings, shields the team from distractions, and ensures adherence to Agile practices.
    - **Development Team**: Self-organizing team responsible for executing the tasks defined in each sprint.
- **Daily Meetings**: Daily "stand-ups" help synchronize the team's progress and discuss any blockers.

Diagram:

- **Sprint Cycle**: The Sprint cycle often includes a flowchart showing the backlog, sprint planning, daily Scrum, sprint review, and sprint retrospective.

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6. Scaling Agile Methods

- **Purpose**: Agile methods are typically suited for small teams. Scaling adapts these methods for larger, multi-team projects.
- **Challenges in Scaling**:
    - **Documentation**: Larger projects require structured documentation to facilitate cross-team collaboration.
    - **Communication**: Clear communication channels, such as regular "Scrum of Scrums" meetings, help maintain coordination.
    - **Continuous Integration**: Scaling agile requires regular integration of work from various teams to ensure a cohesive system.
- **Scaling Strategies**:

- **Scaling Up**: Used when a large, complex system is built by multiple teams.
- **Scaling Out**: Used when agile practices are introduced across an entire organization, impacting multiple departments and workflows.

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7. Agile Development Techniques

- **User Stories**: Short, user-focused descriptions of functionality, helping developers understand customer requirements.
- **Refactoring**: Continuous code improvement. Examples include reorganizing class hierarchies, renaming methods for clarity, and modularizing code.
- **Test-Driven Development (TDD)**: Writing tests before functionality to clarify requirements and maintain high code quality.
- **Automation in Testing**: Tools like JUnit allow for automated testing, ensuring that all new code passes tests for quality control.
- **Pair Programming**: Programmers work in pairs to review each other's code in real-time, sharing knowledge and reducing mistakes.

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8. Practical Challenges in Agile Implementation

- **Contractual Issues**: Traditional contracts focus on fixed deliverables, which may conflict with Agile's iterative nature. Flexible contracts, focusing on developer time rather than specific features, are more compatible with Agile.
- **Maintenance in Agile**: Agile focuses on rapid development, which can challenge long-term maintainability due to minimal documentation. Clear, maintainable code and continued team involvement are crucial.
- **Distributed Teams**: Agile methods often assume a co-located team for efficient communication, but distributed teams require additional tools and processes for collaboration.

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9. Agile Project Management

- **Managing Agile Projects**:

    - Agile project managers facilitate rather than dictate, focusing on removing blockers and empowering the team.
    - Project progress is evaluated continuously, with success measured by working software rather than completion of predefined tasks.
- **Scrum Benefits**:
    - Improved communication, customer visibility, and trust between the development team and customers.
    - Rapid delivery of functional software and adaptability to changing requirements.
    - Better team cohesion and a positive project culture focused on delivering success.

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10. Summary of Agile Principles for Large Systems

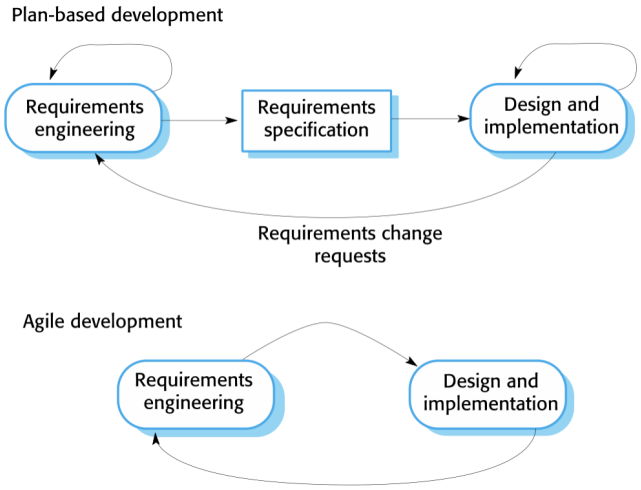
- **Key Points**:
    - Agile methods like Scrum and XP focus on delivering small, usable increments of software and incorporating customer feedback at every stage.
    - Agile practices like user stories, continuous testing, and refactoring support the development of high-quality, adaptable software.
    - Challenges in Agile include scaling, contractual compatibility, maintenance, and documentation needs, especially for large or regulated projects.

Flowchart Overview:

- **Agile Development Process**: Diagrams often show iterative loops involving planning, development, testing, and review stages, highlighting continuous feedback and adaptability.
- **Scaling Agile**: Flowcharts illustrate multi-team integration, communication (e.g., Scrum of Scrums), and the coordination required to maintain a unified project vision across teams.

Plan-driven and agile development



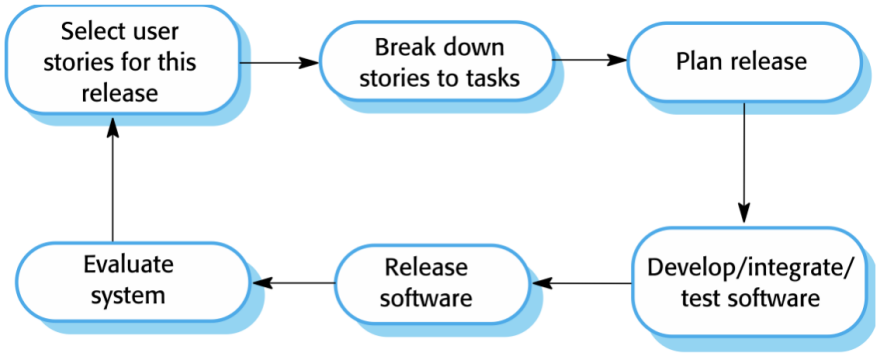


The principles of agile methods



Principle	Description
Customer involvement	Customers should be closely involved throughout the development process. Their role is provide and prioritize new system requirements and to evaluate the iterations of the system.
Incremental delivery	The software is developed in increments with the customer specifying the requirements to be included in each increment.
People not process	The skills of the development team should be recognized and exploited. Team members should be left to develop their own ways of working without prescriptive processes.
Embrace change	Expect the system requirements to change and so design the system to accommodate these changes.
Maintain simplicity	Focus on simplicity in both the software being developed and in the development process. Wherever possible, actively work to eliminate complexity from the system.

The extreme programming release cycle



Extreme programming practices (a)



Principle or practice	Description
Incremental planning	Requirements are recorded on story cards and the stories to be included in a release are determined by the time available and their relative priority. The developers break these stories into development 'Tasks'. See Figures 3.5 and 3.6.
Small releases	The minimal useful set of functionality that provides business value is developed first. Releases of the system are frequent and incrementally add functionality to the first release.
Simple design	Enough design is carried out to meet the current requirements and no more.
Test-first development	An automated unit test framework is used to write tests for a new piece of functionality before that functionality itself is implemented.
Refactoring	All developers are expected to refactor the code continuously as soon as possible code improvements are found. This keeps the code simple and maintainable.

Extreme programming practices (b)



Pair programming	Developers work in pairs, checking each other's work and providing the support to always do a good job.
Collective ownership	The pairs of developers work on all areas of the system, so that no islands of expertise develop and all the developers take responsibility for all of the code. Anyone can change anything.
Continuous integration	As soon as the work on a task is complete, it is integrated into the whole system. After any such integration, all the unit tests in the system must pass.
Sustainable pace	Large amounts of overtime are not considered acceptable as the net effect is often to reduce code quality and medium term productivity
On-site customer	A representative of the end-user of the system (the customer) should be available full time for the use of the XP team. In an extreme programming process, the customer is a member of the development team and is responsible for bringing system requirements to the team for implementation.

Scrum terminology (a)



Scrum term	Definition
Development team	A self-organizing group of software developers, which should be no more than 7 people. They are responsible for developing the software and other essential project documents.
Potentially shippable product increment	The software increment that is delivered from a sprint. The idea is that this should be 'potentially shippable' which means that it is in a finished state and no further work, such as testing, is needed to incorporate it into the final product. In practice, this is not always achievable.
Product backlog	This is a list of 'to do' items which the Scrum team must tackle. They may be feature definitions for the software, software requirements, user stories or descriptions of supplementary tasks that are needed, such as architecture definition or user documentation.
Product owner	An individual (or possibly a small group) whose job is to identify product features or requirements, prioritize these for development and continuously review the product backlog to ensure that the project continues to meet critical business needs. The Product Owner can be a customer but might also be a product manager in a software company or other stakeholder representative.

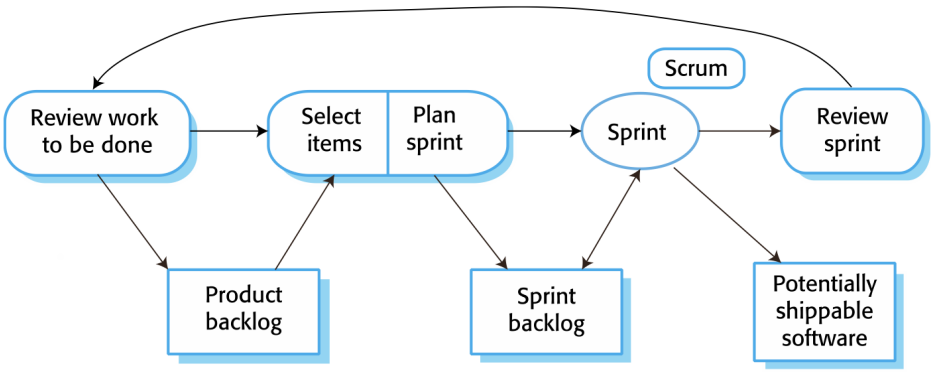
Scrum terminology (b)



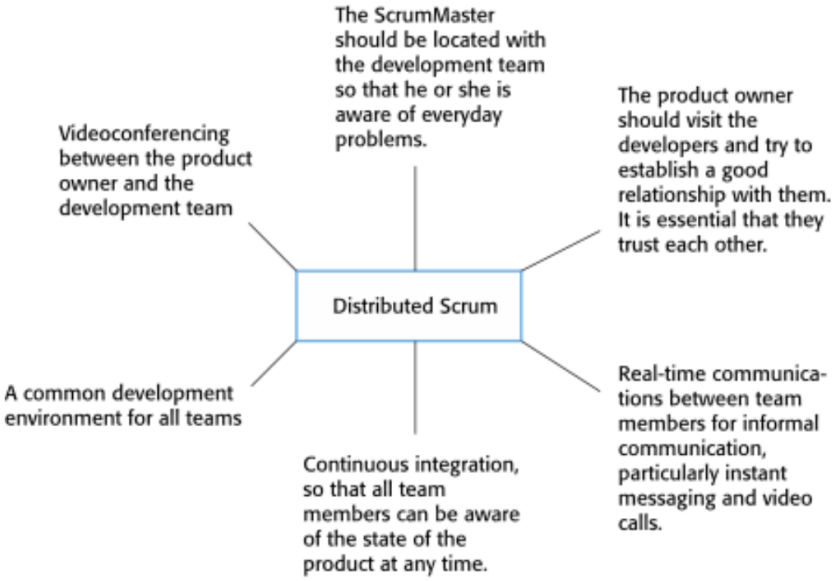
Scrum term	Definition
Scrum	A daily meeting of the Scrum team that reviews progress and prioritizes work to be done that day. Ideally, this should be a short face-to-face meeting that includes the whole team.
ScrumMaster	The ScrumMaster is responsible for ensuring that the Scrum process is followed and guides the team in the effective use of Scrum. He or she is responsible for interfacing with the rest of the company and for ensuring that the Scrum team is not diverted by outside interference. The Scrum developers are adamant that the ScrumMaster should not be thought of as a project manager. Others, however, may not always find it easy to see the difference.
Sprint	A development iteration. Sprints are usually 2-4 weeks long.
Velocity	An estimate of how much product backlog effort that a team can cover in a single sprint. Understanding a team's velocity helps them estimate what can be covered in a sprint and provides a basis for measuring improving performance.

Scrum sprint cycle





Distributed Scrum



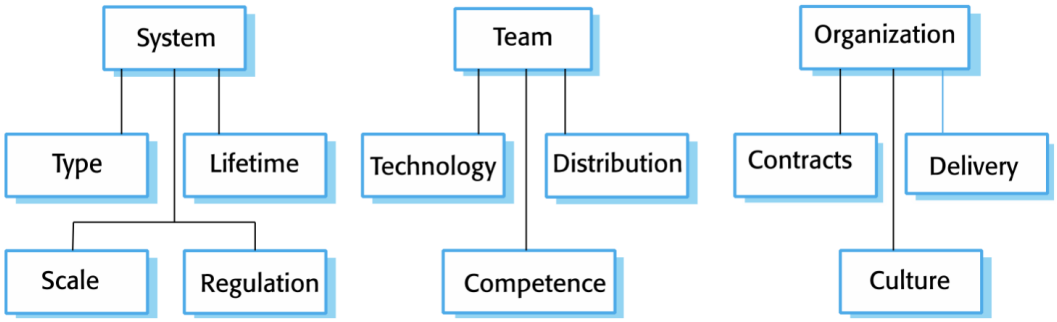
Agile principles and organizational practice

Principle	Practice
Customer involvement	This depends on having a customer who is willing and able to spend time with the development team and who can represent all system stakeholders. Often, customer representatives have other demands on their time and cannot play a full part in the software development. Where there are external stakeholders, such as regulators, it is difficult to represent their views to the agile team.
Embrace change	Prioritizing changes can be extremely difficult, especially in systems for which there are many stakeholders. Typically, each stakeholder gives different priorities to different changes.
Incremental delivery	Rapid iterations and short-term planning for development does not always fit in with the longer-term planning cycles of business planning and marketing. Marketing managers may need to know what product features several months in advance to prepare an effective marketing campaign.

Agile principles and organizational practice

Principle	Practice
Maintain simplicity	Under pressure from delivery schedules, team members may not have time to carry out desirable system simplifications.
People not process	Individual team members may not have suitable personalities for the intense involvement that is typical of agile methods, and therefore may not interact well with other team members.

Agile and plan-based factors



Factors in large systems

